

2 Description

The STM32C53xxx devices are general purpose microcontrollers family (STM32C5 series) based on the high-performance Arm® Cortex®-M33 32-bit RISC core. They operate at a frequency of up to 144 MHz.

The Cortex®-M33 core features a single-precision floating-point unit (FPU) that supports all the Arm® single-precision data-processing instructions and all the data types.

The Cortex®-M33 core also implements a full set of digital signal processing (DSP) instructions and a memory protection unit (MPU) that enhances the application security.

The devices embed high-speed memories (up to 256-Kbyte flash memory and 64-Kbyte SRAM), and an extensive range of enhanced I/Os, peripherals connected to three APB buses, three AHB buses, and a 32-bit multi-AHB bus matrix.

The devices feature several protection mechanisms for embedded flash memory and SRAM: readout protection, write protection, and hide protection areas.

The devices embed several peripherals reinforcing security:

- HASH hardware accelerator
- True random number generator

The devices offer one 12-bit ADC, two DAC channels, two comparators, one operational amplifier, a low-power RTC, one 32-bit general-purpose timer, two 16-bit PWM timers dedicated to motor control, two 16-bit general-purpose timers, two 16-bit basic timers, and one 16-bit low-power timer.

The devices also feature standard and advanced communication interfaces such as:

- One I²C
- One I³C
- Two SPIs with muxed full-duplex I2S
- Two USARTs, two UARTs, and one low-power UART
- Two FDCANs
- One USB full-speed

The devices operate in the -40 to +125 °C (+140 °C junction) temperature ranges from a 2.7 to 3.6 V power supply.

A comprehensive set of power-saving modes allows the design of low-power applications.

The devices offer multiple packages from 20 to 64 pins.

See [Table 1](#) for the list of peripherals available for each part number.

Table 1. Device features and peripheral counts

Peripheral		STM32C53xFxP	STM32C53xFxU	STM32C53xExU	STM32C53xKxT	STM32C53xKxU	STM32C53xCxT/ STM32C53xCxU	STM32C3x2RxT
Flash (Kbytes)		256/128						
SRAM (Kbytes)		64 (including 32 with ECC)						
ICACHE (Kbytes)		8						
Data Flash memory (Kbytes)		64						
One-TimeProgrammable (Kbytes)		4.5						
Timers	Advanced control	2						
	General purpose	1 (32-bit) and 2 (16-bit)						